

Facial evidence matters less for stereotype-consistent judgments:
Weaker formidability-aggression associations in Black vs. White faces

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Abstract

Facial characteristics are reliably associated with social judgments, but these relationships may depend on whether a judgment is consistent with stereotypes of the target's social group. Across four studies with adult participants, we tested this possibility using perceived aggression and threat in Black and White faces. Perceived dominance (Study 1, $N=117$) and physical strength (Study 2, $N=626$) predicted aggression judgments less strongly for Black than White faces. Dominance also predicted perceived threat less strongly for Black than White faces after controlling for several threat-related covariates (Study 3, $N=380$). Furthermore, machine learning analyses found less overlap in dominance- and threat-predictive facial information for Black faces. Finally, face morphing that matched featural variation across race while manipulating apparent race showed a weaker dominance-aggression relationship for Black-appearing faces (Study 4, $N=467$). Together, these findings are consistent with the possibility that facial evidence has reduced influence on judgments of stereotype-consistent traits.

Keywords: faces, formidability, dominance, aggression, stereotype, machine learning

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Facial evidence matters less for stereotype-consistent judgments: Weaker formidability-aggression associations in Black vs. White faces

People form rapid impressions of others' traits based solely on their facial appearance (Willis & Todorov, 2006). Although appearance-based trait impressions are often inaccurate (Jaeger et al., 2020), they can have real and often harmful social consequences, from election outcomes to criminal sentencing (Blair et al., 2004; Eberhardt et al., 2006; Todorov et al., 2005; Wilson & Rule, 2015). For example, how formidable (e.g., physically strong) someone appears is closely associated with judgments of whether a person poses a threat or is likely to behave aggressively (Carré et al., 2009), which may in turn shape perceivers' willingness to approach, avoid, or interact with them.

Previous research has identified robust statistical relationships between facial appearance and specific trait judgments that are consistent across perceivers and cultures (Jones et al., 2021; Oosterhof & Todorov, 2008; Zebrowitz & Montepare, 2008), effectively serving as facial "stereotypes" (Chua & Freeman, 2022). However, recent work has demonstrated that facial impressions often vary based on perceiver and target characteristics (Hehman et al., 2017; Xie et al., 2019). For example, perceivers' beliefs about which traits go together (e.g., are trustworthy people also more confident?) predict how they judge faces (Stolier et al., 2018), relationships between trait judgments differ when viewing physically identical ingroup versus outgroup faces (Hong & Freeman, 2024), and race and gender stereotypes predict how traits are associated with one another in facial impressions (Xie et al., 2021).

Together, this work shows that social expectations can shape impressions formed from faces. However, less is known about how these expectations influence the statistical relationship between facial appearance and trait judgments. One possibility is that this relationship depends

on whether a judgment is consistent with stereotypes of the target's social group. The shifting standards model provides one account of how stereotype consistency might shape this relationship: stereotypes can lead perceivers to evaluate members of different social groups against different standards. Specifically, perceivers require less evidence (i.e., fewer behaviors) to infer stereotype-consistent traits and more evidence to infer stereotype-inconsistent traits. Similar processes may operate when interpreting facial information (e.g., Hess et al., 2004). Applied to facial impressions, perceivers may require less facial evidence to infer traits that are consistent with a target's group stereotypes.

Facial formidability provides a useful test of this possibility. Previous research has identified a robust association between facial formidability cues (e.g., facial width-to-height ratio) and perceived aggression (Carré et al., 2009, 2010; Lefevre & Lewis, 2014). In the United States, Black individuals are often stereotyped as aggressive and dangerous (Devine, 1989). From a shifting standards perspective, aggression may therefore require less facial evidence to be inferred from Black than White faces, resulting in a weaker relationship between facial formidability and perceived aggression for Black faces. A competing account, however, predicts the opposite pattern. Racial stereotypes may heighten sensitivity to stereotype-consistent information (Allen et al., 2009; Wigboldus et al., 2003), increasing rather than decreasing the extent to which perceivers rely on stereotype-consistent facial information. Previous research has documented a number of stereotype-consistent biases in threat perception: Black men are perceived as physically larger and more threatening than White men (Wilson et al., 2017), and perceivers higher in implicit prejudice perceive anger more readily and intensely in Black than White faces (Hugenberg & Bodenhausen, 2003; Hutchings & Haddock, 2008). Further, non-facial dominance cues such as height and physical size have been shown to increase perceptions

of threat for Black targets (Hester & Gray, 2018). Thus, rather than reducing the importance of formidability information, racial stereotypes may amplify its association with aggression for Black targets.

To adjudicate between these possibilities, we conducted four studies examining whether facial formidability is differentially associated with perceived aggression and threat in Black and White faces. In Study 1, we examined whether the relationship between perceived dominance and aggression differs between Black and White male faces. In Study 2, we replicated this effect using perceived physical strength as an alternative measure of formidability and included both male and female faces. In Study 3, we tested whether the racial difference in the dominance-threat relationship persists after controlling for other facial characteristics associated with threat judgments, and whether the facial information associated with dominance and threat judgments overlaps differently for Black and White faces. Finally, Study 4 tested whether the relationship tracks targets' apparent race rather than differences in facial structure between Black and White faces.

Research Transparency Statement

General Disclosures

Conflicts of interest: All authors declare no conflicts of interest. **Funding:** We received no funding for this research. **Artificial intelligence:** ChatGPT was used to assist with writing code for data cleaning and visualization in R. All code was reviewed by all members of the research team for accuracy. No other AI-assisted technologies were used in this research or the creation of this article. **Ethics:** This research received approval from the Columbia University Institutional Review Board (ID: IRB-AAAU0121) and the University of Colorado Boulder Institutional Review Board (ID: 25-0372).

Study One

Preregistration: No aspects of the study were preregistered. **Materials:** The survey questions are publicly available (<https://researchbox.org/9817> CODE: EZYJEY). Study 1 and all subsequent studies used Chicago Face Database images (CFD; Ma et al., 2015), which cannot be shared because of copyright restrictions imposed by the University of Chicago Center for Decision Research. However, the face images are publicly available for download from the CFD website (<https://www.chicagofaces.org/>). **Data:** All processed data are publicly available (<https://researchbox.org/9817> CODE: EZYJEY). The original raw data for Study 1 were not retained and are no longer available to the authors. **Analysis scripts:** All analysis scripts are publicly available (<https://researchbox.org/9817> CODE: EZYJEY).

Study Two

Preregistration: No aspects of the study were preregistered. **Materials:** The survey questions are publicly available (<https://researchbox.org/9817> CODE: EZYJEY). Face images cannot be shared due to copyright restrictions but are publicly available for download from the CFD website (<https://www.chicagofaces.org/>). **Data:** All processed data are publicly available (<https://researchbox.org/9817> CODE: EZYJEY). The original raw data for Study 2 were not retained and are no longer available to the authors. **Analysis scripts:** All analysis scripts are publicly available (<https://researchbox.org/9817> CODE: EZYJEY).

Study Three

Preregistration: No aspects of the study were preregistered. **Materials:** No study materials are available as this study reanalyzes a publicly available dataset. **Data:** The norming data and face images cannot be shared due to copyright restrictions but are publicly available for

download from the CFD website (<https://www.chicagofaces.org/>). **Analysis scripts:** All analysis scripts are publicly available (<https://researchbox.org/9817> CODE: EYZJEY).

Study Four

Preregistration: No aspects of the study were preregistered. **Materials:** The survey questions and doppelgänger face images are publicly available (<https://researchbox.org/9817> CODE: EYZJEY). Original face images cannot be shared due to copyright restrictions but are publicly available for download from the CFD website (<https://www.chicagofaces.org/>).

Data: All primary data are publicly available (<https://researchbox.org/9817> CODE: EYZJEY).

Analysis scripts: All analysis scripts are publicly available (<https://researchbox.org/9817> CODE: EYZJEY).

Study 1

Study 1 tested whether the relationship between perceived dominance and aggressiveness varies between Black and White target faces. Prior work shows that facial dominance cues are reliably associated with perceived aggression (Carré et al., 2010), and we therefore expect overall strong associations between the two judgments. However, competing accounts suggest that this relationship may differ for Black and White targets. Heightened sensitivity to stereotype-consistent information may strengthen the dominance-aggression association for Black targets, whereas shifting standards may weaken this association if less evidence is required to infer stereotype-consistent aggression.

Methods

Participants. An a priori power analysis was conducted based on the effect size reported in Study 1 of Xie et al. (2021). Specifically, we used the approximated correlation between dominance and aggressiveness judgments for Black and White targets ($r = .668$) as the

benchmark effect size, which corresponded to a target sample size of approximately $N = 18$ for 80% power at $\alpha = .05$. We approximated this correlation because the publicly available dataset only contained Fisher's z-transformed correlations. Because we anticipated possible racial ingroup/outgroup effects in addition to racial category effects, we conservatively quadrupled the target sample size ($N = 80$). To maintain balanced representation across racial groups, we recruited a convenience sample of 120 participants (60 Black participants and 60 White participants) from CloudResearch. Three participants were excluded because their self-reported race in the demographic survey did not match their prescreened race (i.e., participants prescreened as White who later reported a different racial identity). No procedures were implemented to detect AI-generated responses because the study was conducted in 2021 prior to the public availability of generative AI tools such as ChatGPT/Gemini. The final sample consisted of 117 participants ($M_{\text{age}} = 38.98$, $SD_{\text{age}} = 10.79$; 69 female, 48 male; 60 Black, 57 White).

Procedure. Our stimulus set consisted of 40 Black male and 40 White male faces with neutral expressions from the CFD (Ma et al., 2015). Each participant rated 20 unique faces (10 Black, 10 White), randomly selected from the pool of 80 faces. Participants then simply rated the 20 faces on both dominance and aggressiveness. We presented a face image with a prompt (e.g., "How dominant is this person?"). Participants responded using a 7-point Likert scale (1 – not at all, 7 – very much). Participants rated each trait in a block before moving onto the next trait. The order of trait presentation was randomized across participants, and the order of face presentation was randomized across participants and across blocks.

Results

We tested whether the association between perceived dominance and aggressiveness varied by target race and racial ingroup/outgroup status using a linear mixed-effects model predicting aggressiveness from mean-centered dominance, target race (Black = +0.5, White = -0.5), ingroup/outgroup status (ingroup = +0.5, outgroup = -0.5), and their interactions. Random intercepts were included for participants and faces, with random slopes for target race by participant. Ingroup/outgroup status and its interactions did not improve model fit, $\chi^2(4) = 0.93$, $p = .92$, and were removed from the final model.

In the final model, dominance positively predicted aggressiveness, $b = 0.23$, $SE = 0.02$, $t(2261) = 11.73$, $p < .001$, 95% CI [0.19, 0.26]. Target race was not significant, $b = 0.12$, $SE = 0.10$, $t(91) = 1.17$, $p = .245$, 95% CI [-0.08, 0.33], indicating that aggressiveness ratings did not significantly differ between Black targets ($M = 4.35$, $SD = 1.72$) and White targets ($M = 4.22$, $SD = 1.77$). Critically, the dominance by target race interaction was significant, $b = -0.09$, $SE = 0.04$, $t(1988) = -2.35$, $p = .019$, 95% CI [-0.16, -0.01]. Simple slope analyses showed that dominance positively predicted aggressiveness for both White targets, $b = 0.27$, $SE = 0.03$, $p < .001$, and Black targets, $b = 0.18$, $SE = 0.03$, $p < .001$, with a weaker association for Black targets (Figure 1a).

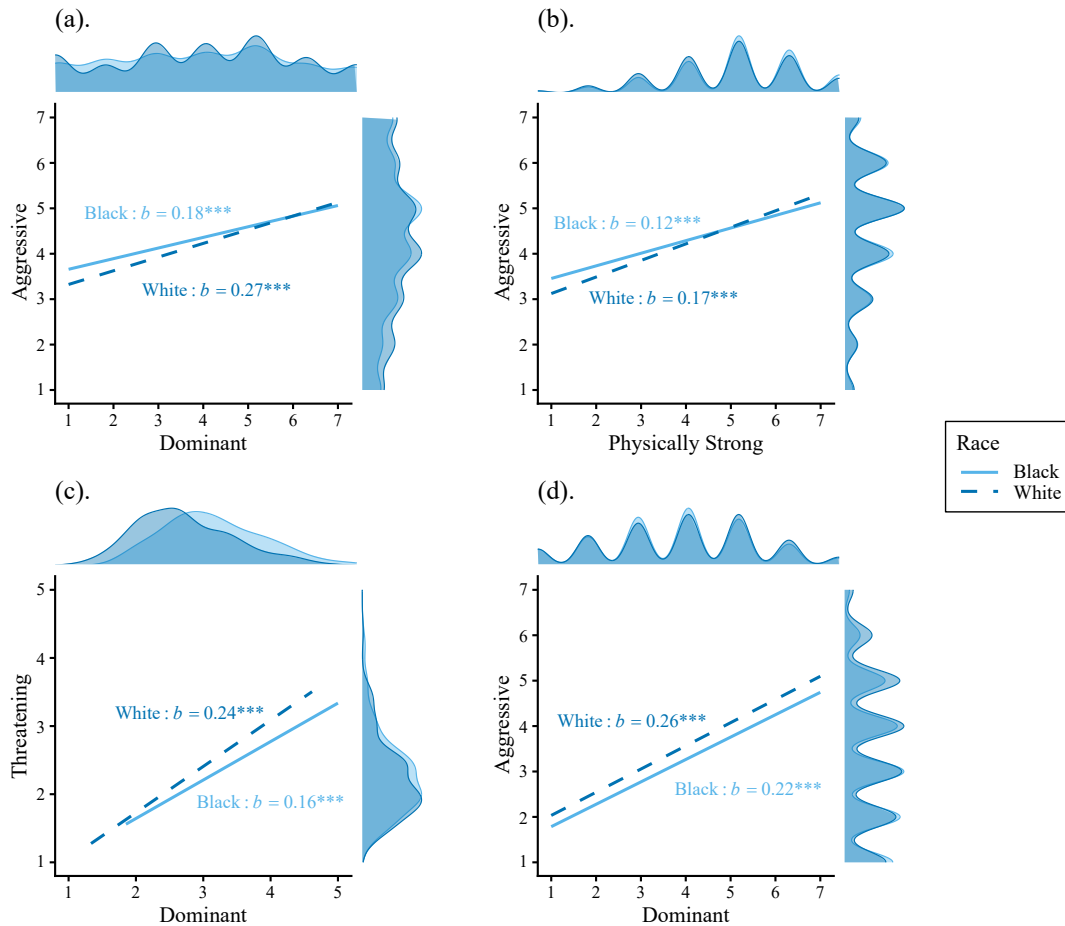


Figure 1. Relationships between facial formidability and perceived aggression and threat by target race. Each panel shows regression lines separately for Black and White targets. Relationship between: (a) Study 1: perceived dominance and aggressiveness. (b) Study 2: perceived physical strength and aggressiveness. (c) Study 3: perceived dominance and threat. (d) Study 4: perceived dominance and aggressiveness. Across all studies, the relationship between facial formidability and aggression/threat judgments was significantly weaker for Black than for White targets. The distributions for (a), (b), and (d) include by-image ratings for each participant and are thus more jagged than the distribution for (c), which includes by-image ratings averaged across CFD raters.

Discussion

Study 1 showed that perceived dominance was positively associated with perceived aggression across both Black and White targets, and this association was weaker for Black than White targets, consistent with the shifting-standards prediction that less facial evidence may be required to infer stereotype-consistent aggression for Black targets. Notably, this effect was not

driven by a simple main effect of target race, as overall aggressiveness ratings did not differ significantly between Black and White faces. The weaker dominance-aggressiveness association for Black targets was also consistent across racial ingroup and outgroup targets.

Study 2

Study 2 replicated Study 1 using an alternative measure of facial formidability: perceived physical strength. We also tested whether effects generalized across target gender by including both male and female faces.

Methods

Participants. An a priori power analysis for the multiple regression was again conducted based on Study 1 of Xie et al. (2021). We used the approximated correlation between judgments of physical strength and aggression for Black and White targets ($r = .299$) as the benchmark effect size, which yielded a required sample size of approximately $N = 88$ to achieve 80% power at $\alpha = .05$. To prevent potential confounds related to participant race, recruitment was limited to individuals identifying as Black or White, as in Study 1. We recruited a convenience sample on CloudResearch; data collection continued until we reached the target range, resulting in 630 participants.¹ No procedures were implemented to detect AI-generated responses because the study was conducted in 2021 prior to the public availability of generative AI tools such as ChatGPT/Gemini. After excluding four participants due to discrepant racial reporting between prescreen and end-of-study demographics, the final sample consisted of 626 participants ($M_{\text{age}} = 36.96$, $SD_{\text{age}} = 10.31$; 293 female, 333 male; 240 Black, 386 White).

Procedure. The Study 2 procedure largely followed that of Study 1 with a few exceptions. We expanded the stimulus set to include both male and female faces to examine

¹ An earlier version of the power analysis containing a technical error yielded a required sample of $N = 612$. The error was discovered after data collection; thus, the sample is larger than required.

whether the effects generalized across target gender (80 Black faces and 80 White faces, with equal numbers of male and female identities within each race category). Faces were again drawn from the CFD. Each participant rated 20 unique faces (10 Black and 10 White faces) of one gender, randomly selected from the full stimulus pool. Participants rated each face on perceived physical strength and aggressiveness. On each trial, a face image was presented with a prompt (e.g., “How physically strong is this person?”), and responses were made on a 7-point Likert scale (1 = not at all, 7 = very much). Participants completed ratings for one trait in a block before proceeding to the next trait. The order of trait blocks was randomized across participants, and face order was randomized both within and across trait blocks.

Results

We tested whether the association between perceived physical strength and aggressiveness varied by target race and ingroup/outgroup status using a linear mixed-effects model predicting aggressiveness from mean-centered perceived physical strength, target race (Black = +0.5, White = -0.5), ingroup/outgroup status (ingroup = +0.5, outgroup = -0.5), and their interactions. Random intercepts were included for participants and faces, with random slopes for target race by participant. Ingroup/outgroup status and its interactions did not improve model fit, $\chi^2(4) = 3.04$, $p = .551$, and were removed from the final model.

In the final model, perceived physical strength significantly predicted aggressiveness, $b = 0.15$, $SE = 0.01$, $t(12442) = 15.55$, $p < .001$, 95% CI [0.13, 0.16]. The main effect of target race was not significant, $b = 0.03$, $SE = 0.04$, $t(256) = 0.66$, $p = .508$, 95% CI [-0.06, 0.11], indicating that aggressiveness ratings did not differ between Black targets ($M = 4.55$, $SD = 1.46$) and White targets ($M = 4.50$, $SD = 1.48$). Critically, the perceived physical strength by target race interaction was significant, $b = -0.05$, $SE = 0.02$, $t(8226) = -2.61$, $p = .009$, 95% CI [-0.08, -0.01],

indicating that the association between perceived physical strength and aggressiveness differed by target race. Simple slope analyses showed that perceived physical strength positively predicted aggressiveness for both White targets, $b = 0.17$, $SE = 0.01$, $p < .001$, and Black targets, $b = 0.12$, $SE = 0.01$, $p < .001$, with a weaker association for Black targets (Figure 1b).

Discussion

Study 2 showed that perceived physical strength was positively associated with aggressiveness judgments across targets, and this association was again weaker for Black than White targets.

Study 3

Study 3 tested whether the weaker formidability-threat relationship for Black faces persisted after controlling for several potentially relevant covariates, such as perceived anger and facial masculinity/femininity. We also asked whether the racial difference observed in trait judgments was reflected in the facial information associated with those judgments. Using data-driven image analysis, we identified facial information predictive of perceived dominance and threat and tested whether this information overlapped less for Black than White faces.

Methods

Stimuli. We analyzed all 380 Black and White face images available in the CFD along with their associated norming data. One White male face was excluded from the machine learning analyses because it could not be successfully pre-processed using OpenFace. The stimulus set included 104 Black female, 93 Black male, 90 White female, and 93 White male faces. Because the CFD norming data do not include aggressiveness ratings, we used perceived threat as a closely related outcome measure (Wilson et al., 2017). In addition to dominance and threat ratings, the CFD norming dataset includes several variables that may be associated with

facial threat perception, including racial prototypicality, proportion of raters categorizing the face as White or Black, anger, facial masculinity, facial femininity, and facial width-to-height ratio (Adams et al., 2022; Geniole et al., 2015; Hugenberg & Bodenhausen, 2004; Lefevre & Lewis, 2014). We therefore included these variables as covariates in our analyses to account for alternative sources of variance in threat-related judgments. Sample size was constrained by the number of available face images in the database; sensitivity analyses are reported in the Results section. The number of raters per face ranged from 21-131 ($M = 53.63$, $SD = 32.07$).

Results

Face ratings. We tested whether the association between perceived dominance and threat varied by target race after controlling for several facial and perceptual covariates, including perceived anger, femininity, masculinity, racial prototypicality, face width-to-height ratio, and Black and White categorization probabilities. Threat was regressed on mean-centered perceived dominance, target race (Black = +0.5, White = -0.5), their interaction, and covariates. Perceived dominance significantly predicted perceived threat, $b = 0.20$, $SE = 0.03$, $t(369) = 6.98$, $p < .001$, 95% CI [0.15, 0.26], indicating that more dominant-looking faces were judged as more threatening. There was also a significant main effect of target race, $b = -0.22$, $SE = 0.10$, $t(369) = -2.24$, $p = .026$, 95% CI [-0.42, -0.03], indicating that White faces were rated as more threatening than Black faces after accounting for facial cues and other covariates (covariate-adjusted marginal means for White faces: $M = 2.37$, $SE = 0.05$, 95% CI [2.26, 2.47] vs. Black faces: $M = 2.14$, $SE = 0.05$, 95% CI [2.05, 2.24]). Critically, this relationship was qualified by a significant dominance by target race interaction, $b = -0.08$, $SE = 0.04$, $t(369) = -2.29$, $p = .023$, 95% CI [-0.15, -0.01]. Simple slope analyses indicated that perceived dominance positively predicted threat for both Black and White faces, but the association was weaker for Black faces ($b = 0.16$,

$SE = 0.03, p < .001$) than for White faces ($b = 0.24, SE = 0.04, p < .001$) (Figure 1c). A sensitivity analysis implemented in R indicated that this sample size ($N = 380$ images) provided 80% power to detect an interaction between race and dominance with an effect size equivalent to $|b| = 0.10$ or greater, which is slightly larger than the observed effect size of $b = -0.08$.

Face image analysis. To examine how facial information relates to dominance and threat judgments across racial groups, we conducted a data-driven image analysis using machine learning approaches (Hong et al., 2023). Inner face regions were first extracted from each image using OpenFace (Baltrušaitis et al., 2016), and images were then spatially normalized such that key facial landmarks (eyes, nose, and mouth) were aligned across stimuli. Given the computational demands of high-dimensional image analysis, images were down-sampled from 512x512 to 128x128 pixels and standardized prior to analysis. Prior work using comparable procedures has shown that this level of downsampling does not meaningfully reduce predictive performance (Hong & Ratner, 2021). We then used support vector regression (SVR; Drucker et al., 1997) with a linear kernel to predict continuous trait judgments from pixel-level facial information within each race-by-trait condition (Black dominant, Black threatening, White dominant, White threatening). SVR is well suited for this setting, where the number of predictors (4,096 pixels) far exceeds the number of observations ($n = 197$ for Black, $n = 182$ for White targets). We used the default cost parameter ($C = 1$) and did not perform hyperparameter tuning, as the goal was not to optimize absolute predictive accuracy but to characterize and compare predictive patterns across race-by-trait conditions.

Model performance was evaluated using 10-fold cross-validation. In each fold, models were trained on 90% of the images and tested on the remaining 10%, ensuring that each image served as a test case exactly once. Predictive accuracy was quantified using mean absolute error

(MAE), with lower values indicating better fit between predicted and observed ratings. To assess whether prediction performance exceeded chance, we conducted 1,000 permutation tests separately for each condition (Ojala & Garriga, 2010). In each iteration, trait ratings were randomly shuffled across images, breaking any true association between facial structure and judgments while preserving the distribution of the data. The full cross-validation procedure was then repeated on the permuted datasets. Statistical significance was determined by comparing observed MAE values to the permutation distribution. Across all conditions, prediction accuracy significantly exceeded chance, indicating that pixel-level facial information reliably predicted both dominance and threat judgments for Black and White faces (Black dominant MAE = .50, $SD = .05$, $p < .001$; Black threatening MAE = .43, $SD = .07$, $p = .002$; White dominant MAE = .49, $SD = .08$, $p < .001$; White threatening MAE = .42, $SD = .04$, $p < .001$). Permutation test results are shown in Figure 2.

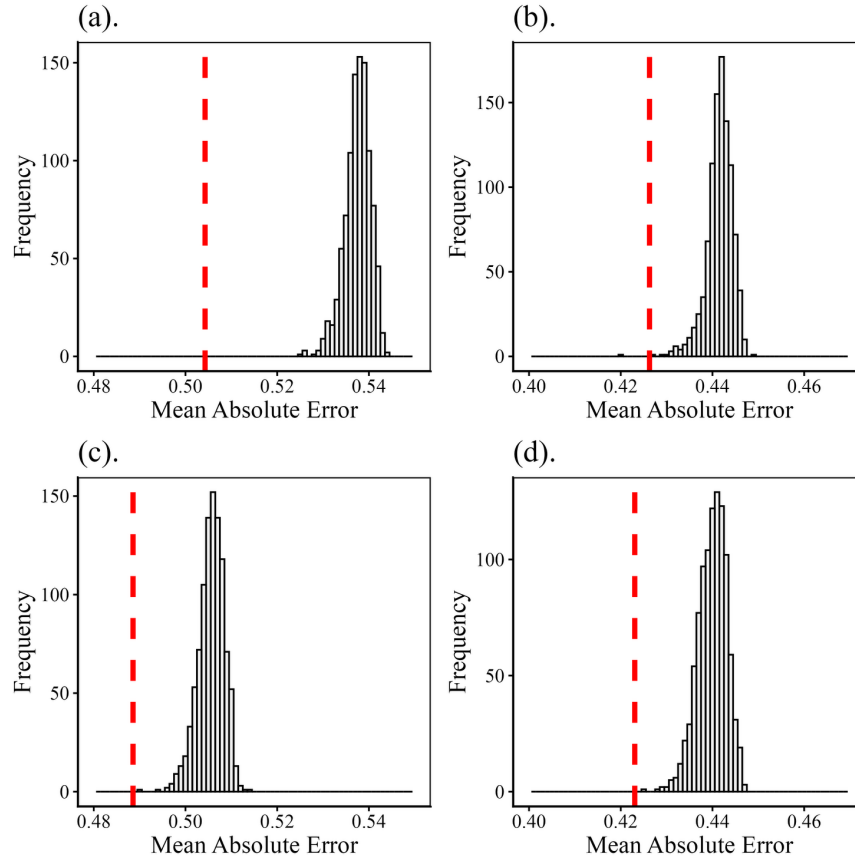


Figure 2. Permutation test results for trait predictions: (a) Black dominant, (b) Black threatening, (c) White dominant, and (d) White threatening. The dashed lines indicate true MAE scores. All predictions were statistically significant ($ps < .01$).

To identify the facial information most predictive of dominance and threat judgments, we performed a pixelwise variable-ranking procedure (Guyon & Elisseeff, 2003). Whereas SVR tests whether the joint pixel pattern predicts ratings, variable ranking characterizes where in the face rating-associated information is located at the pixel level. Importance was estimated separately for each retained pixel and scaled from 0 to 100, with larger values indicating that variation in a pixel's intensity was more predictive of variation in ratings. The resulting maps were overlaid on a sample face image to display the spatial distribution of judgment-associated facial information within each condition (Figure 3a).

Lastly, we quantified overlap between importance maps using Euclidean distance between maps, with smaller values indicating greater similarity. Because neighboring pixels are spatially dependent (Petrov & Zhaoping, 2003), confidence intervals were estimated using a spatial block bootstrap (Hall, 1985; Lahiri, 2003). Across 1,000 iterations, 8x8-pixel blocks were sampled with replacement, with identical block locations used across maps within each iteration.

Dominance and threat maps showed significant overlap for both Black and White targets. Observed distances were 0.72, 95% CI [0.66, 0.79] for Black targets and 0.55, 95% CI [0.49, 0.60] for White targets, both significantly smaller than their respective permutation-null means of 1.40 and 1.36 ($ps < .001$). However, the maps were significantly more similar for White than Black targets, difference = 0.18, 95% CI [0.10, 0.25], $p < .001$. Because dominance and threat ratings were also more strongly correlated for White targets, we tested whether this difference reflected rating-level covariation. Threat ratings were regressed on dominance ratings at the image level, and threat maps were recomputed from the residuals. After removing this association, distances were 1.38, 95% CI [1.30, 1.45], for Black targets and 1.25, 95% CI [1.17, 1.32], for White targets. The White-target maps remained more similar, difference = 0.13, 95% CI [0.03, 0.21], $p = .010$. Thus, the greater similarity between dominance- and threat-predictive facial information for White targets could not be explained by the stronger correlation between dominance and threat ratings for White targets.

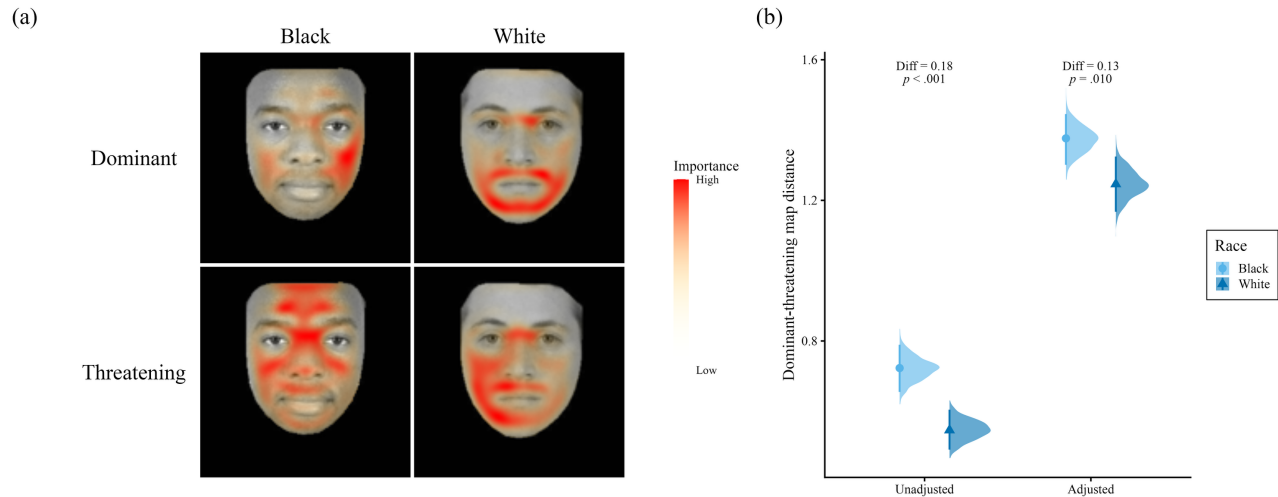


Figure 3. (a) Facial importance maps for dominance and threat ratings of Black and White targets. (b) Distance between dominance and threat importance maps, before and after adjusting for rating correlations. Larger values indicate that the maps are less similar. Points show observed differences with 95% CIs, and densities show block-bootstrap distributions.

Discussion

Study 3 showed that perceived dominance predicted perceived threat, and this relationship was again weaker for Black than White faces, even after controlling for several facial and perceptual covariates associated with threat-related judgments. Moreover, machine learning analyses showed less overlap between the facial information predictive of dominance and threat judgment for Black than White faces, paralleling the trait rating results.

Study 4

The observed asymmetry in the relationship between facial formidability and perceived aggression and threat for Black and White faces may reflect naturally occurring differences in facial structure between the Black and White faces (Adams et al., 2022; Zebrowitz et al., 2010). Study 4 addressed this possibility by manipulating targets' apparent race while matching facial variation across racial categories. We then tested whether the weaker formidability-aggression

association for Black faces would follow targets' apparent race even when facial variation was matched.

Methods

Participants. We used the R package *mixedpower* (version 0.1.0; Kumle et al., 2021) to conduct an a priori simulation power analysis based on the data and mixed-effects model from Study 1. The simulation indicated that a sample size of $N = 200$ would provide 87.2% power to detect an interaction between target race and perceived facial dominance on perceived aggressiveness, using the effect size for this interaction from Study 1 as a benchmark and a significance threshold of $\alpha = .05$. We aimed to collect approximately 25 participant ratings per face image (704 images total) and asked each participant to rate only 40 images to avoid fatigue and inattentive responding. These constraints produced a target sample size of 438 participants. To allow for exclusions, we recruited a convenience sample of 500 participants on Prolific. Recruitment was limited to individuals identifying as Black or White, as in Studies 1-2. Data collection continued until we reached the target range, resulting in 502 participants. After excluding 28 participants due to discrepant racial reporting between prescreen and end-of-study demographics, the final sample consisted of 474 participants ($M_{\text{age}} = 41.58$, $SD_{\text{age}} = 10.93$; 249 women, 220 men, 5 nonbinary/other; 75 Black, 399 White). To detect AI-generated responses, we implemented a CAPTCHA via Qualtrics which produces scores ranging from 0-1. While there is no established threshold of acceptability for these scores, Qualtrics documentation suggests that a score below .5 may indicate bot-generated responses. Of our 474 participants, 17 either had a score below .5 or had no score recorded due to technical errors (e.g., ad blockers or firewalls). The reported results do not change in direction or significance when these participants are excluded (see Supplemental Materials), so we retain them in our sample.

Stimuli. The stimulus set included 352 original face images (94 Black female, 89 Black male, 86 White female, and 83 White male) drawn from the CFD, as well as their other-race “doppelgängers” (Black faces altered to appear White and vice versa). The final stimulus set thus contained 704 total faces. Doppelgänger faces were created using Psychomorph software (Tiddeman et al., 2001), which allowed us to preserve the distinctive pattern of facial variation associated with a particular individual while altering category-diagnostic racial information (Figure 4). For each face, we characterized its pattern of shape, texture, and color deviations from the average face of the same race and gender and applied this pattern to the corresponding average face of the other race. This procedure generated Black- and White-appearing doppelgängers that shared the same pattern of facial variation relative to their respective race-by-gender category averages (Cohen et al., 2026). The procedure was applied in both directions, such that Black faces were transformed to appear White and White faces were transformed to appear Black, while gender was held constant.

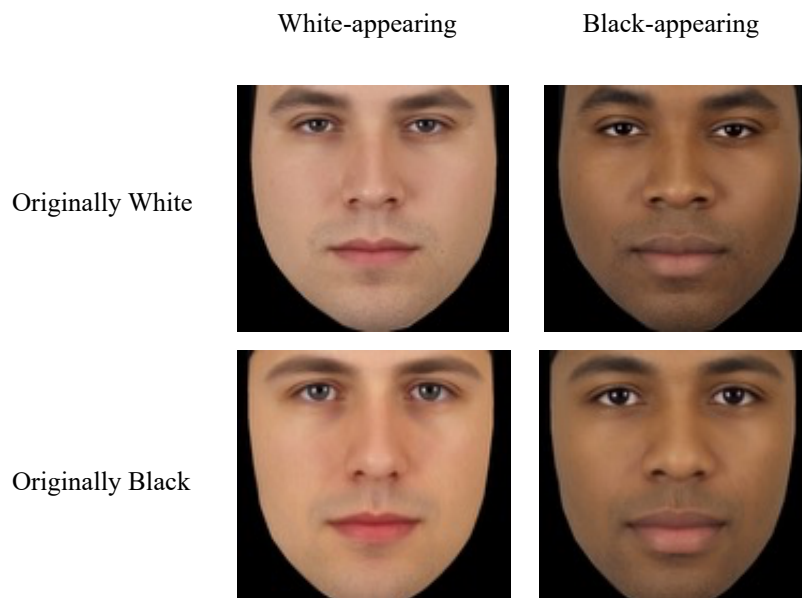


Figure 4. Sample doppelgänger stimuli. The morphing procedure preserved each target’s unique facial appearance while altering their apparent race.

Procedure. The Study 4 procedure largely followed that of Studies 1-2 with a few differences. Each participant rated 40 unique faces (10 Black original, 10 Black doppelgängers, 10 White original, and 10 White doppelgängers) of one gender, pseudorandomly selected from the full stimulus pool such that participants might rate either the original or doppelgänger version of a face identity, but never both. Participants rated each face on perceived dominance and aggressiveness. On each trial, a face image was presented with a prompt (e.g., “How dominant is this person?”), and responses were made on a 7-point Likert scale (1 – not at all, 7 – very much). Participants completed ratings for one trait in a block before proceeding to the next trait. The order of trait blocks was randomized across participants, and face order was randomized both within and across trait blocks.

Results

We tested whether the association between dominance and aggressiveness varied by target race and original/doppelgänger status using a linear mixed-effects model predicting aggressiveness from mean-centered perceived dominance, target apparent race (Black-appearing = +0.5, White-appearing = -0.5), original/doppelgänger status (original = +0.5, doppelgänger = -0.5), and their interactions. Random intercepts were included for participants and target identity (i.e., the shared identity of each original/doppelgänger pair), with random slopes for target race by participant.

Perceived dominance significantly predicted perceived aggressiveness, $b = 0.24$, $SE = 0.01$, $t(18894) = 36.55$, $p < .001$, 95% CI [0.23, 0.26]. Target apparent race also significantly predicted perceived aggressiveness, $b = -0.32$, $SE = 0.03$, $t(473) = -10.80$, $p < .001$, 95% CI [-0.38, -0.26], indicating that aggressiveness ratings were lower for Black-appearing targets ($M = 3.17$, $SD = 1.56$) than for White-appearing targets ($M = 3.51$, $SD = 1.64$). The main effect of

target apparent race was qualified by a significant interaction between perceived dominance and target apparent race, $b = -0.04$, $SE = 0.01$, $t(13768) = -3.28$, $p = .001$, 95% CI $[-0.06, -0.02]$, indicating that the association between perceived dominance and aggressiveness differed by target apparent race. Simple slope analyses showed that perceived dominance positively predicted aggressiveness for both White-appearing targets, $b = 0.26$, $SE = 0.01$, $p < .001$, and Black-appearing targets, $b = 0.22$, $SE = 0.01$, $p < .001$, with a weaker association for Black-appearing targets (Figure 1d).

Finally, original/doppelgänger status significantly predicted perceived aggressiveness, $b = -0.04$, $SE = 0.01$, $t(17694) = -2.89$, $p = .004$, 95% CI $[-0.07, -0.01]$, indicating that aggressiveness ratings were lower for original targets ($M = 3.31$, $SD = 1.60$) than for doppelgänger targets ($M = 3.37$, $SD = 1.61$). This main effect was qualified by an interaction with target apparent race, $b = 0.23$, $SE = 0.10$, $t(336) = 2.29$, $p = .02$, 95% CI $[0.03, 0.42]$, such that White-appearing doppelgängers ($M = 3.60$, $SD = 1.64$) were rated as more aggressive than original White targets ($M = 3.41$, $SD = 1.63$), but there was no significant difference in aggressiveness ratings between Black-appearing doppelgängers ($M = 3.14$, $SD = 1.54$) and original Black targets ($M = 3.21$, $SD = 1.57$). There was not a significant three-way interaction between original/doppelgänger status, perceived dominance, and target race, $p > .05$.

Discussion

Study 4 showed that the association between perceived dominance and aggressiveness was again weaker for Black than White targets. This effect followed targets' apparent race, regardless of whether the face was an original or a race-morphed version. By matching patterns of facial variation across Black- and White-appearing targets, Study 4 reduced the possibility that

the racial difference observed in Studies 1-3 reflected differences in facial structure between Black and White faces.

General Discussion

Across four studies, facial formidability was consistently less strongly associated with perceived aggression and threat for Black than White faces. This pattern emerged across perceived dominance and physical strength (Studies 1-2), persisted after accounting for other measured facial and perceptual characteristics, and was reflected in less overlap between dominance- and threat-predictive facial information (Study 3). Finally, Study 4 showed that the weaker dominance-aggression association followed targets' apparent race when facial variation was matched across racial categories. Together, these findings demonstrate a robust racial difference in the relationship between facial formidability and perceived aggression.

More broadly, these results suggest that social knowledge, such as stereotypes linking Black individuals with aggression (Devine, 1989), may influence otherwise robust relationships between facial appearance and trait judgments. The weaker relationship for Black faces is consistent with the shifting standards model (Biernat & Ma, 2005), whereby less evidence is required to infer stereotype-consistent traits. At first glance, this pattern appears to contrast with evidence that racial stereotypes can amplify threat-related perceptions of Black targets (Hugenberg & Bodenhausen, 2003). However, stereotype-consistent amplification and shifting standards may reflect distinct consequences of social knowledge for facial impressions. Racial stereotypes may bias ambiguous or stereotype-relevant information toward threat (Sagar & Schofield, 1980) while simultaneously reducing the amount of additional facial evidence required for an aggression-related inference. Thus, greater stereotype-consistent bias need not produce a stronger relationship between formidability and aggression or threat. Although the

present studies do not directly test this process, this possibility could reconcile our findings with previous evidence of stereotype-consistent amplification.

The present findings also have implications for understanding appearance-based social judgments. Research on face perception has identified robust relationships between facial characteristics and trait impressions (Oosterhof & Todorov, 2008; Zebrowitz & Montepare, 2008). Our findings demonstrate that even the well-established relationship between facial formidability and perceived aggression is not invariant across social groups. Social knowledge may therefore shape not only mean differences in impressions, but also how strongly perceived facial characteristics relate to these judgments. This distinction is important because Black faces were never rated as more aggressive or threatening than White faces across our studies, yet race consistently moderated the relationship between facial formidability and aggression-related judgments. Thus, the absence of a stereotypical mean difference does not imply that social knowledge is irrelevant to how impressions are formed.

An interesting future question is which mechanisms produce this difference. One possibility is that racial category changes the meaning of facial information when forming social judgments. For example, perceived untrustworthiness is associated with different emotion-resembling facial information across racial groups, suggesting that similar social judgments can arise from different facial information depending on target race (Klein et al., 2026). In the present context, facial formidability may primarily signal capacity to cause harm, whereas aggression and threat judgments may additionally involve inferences about intent to cause harm (MacDonell et al., 2018). In our studies, formidability might have signaled capacity for both Black and White targets. However, if racial stereotypes make aggressive intent more readily inferred for Black targets, judgments of their aggression may depend less on variation in

perceived capacity. Future studies could separately measure or manipulate facial formidability and aggressive intent to test this possibility.

Despite its contributions, this work has several limitations. First, we focused on Black and White targets and aggression and threat, a domain in which racial stereotypes are well documented. It remains unclear whether the same pattern extends to other social groups and stereotyped traits, particularly stereotypes that differ in strength or content. Testing multiple groups and stereotype-consistent, stereotype-inconsistent, and stereotype-irrelevant traits would help establish the generality and boundary conditions of the effect. Second, we inferred the role of racial stereotypes from established cultural associations rather than directly measuring participants' stereotype beliefs. Future work should test whether individual differences in knowledge or endorsement of these stereotypes predict the strength of the formidability-aggression relationship (see Xie et al., 2021).

The present work demonstrates that a well-established relationship between facial appearance and social judgment is not constant across social groups. Facial formidability was less strongly associated with aggression and threat for Black than White faces, and this difference followed targets' apparent race even when facial variation was matched. These findings demonstrate that understanding bias in face perception requires considering not only whether social groups are judged differently, but whether the same perceived facial characteristics relate differently to social judgments across groups.

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